

Institute of Actuaries of India

Subject ST5 – Finance and Investment A

March 2018 Examination

INDICATIVE SOLUTION

Solution 1:

Asset liability mismatch reserving involves projecting the emerging asset and liability position under a range of possible conditions in order to establish the extent to which assets and liabilities are mismatched. Appropriate supplementary reserves can then be set up to cover the possible levels of shortfall identified.

Using stochastic techniques, multiple projections would be made in order to generate many possible future scenarios.

Most often, the stochastic element of the projections would apply to the asset portfolio and investment returns, in order to assess exposure to systematic risk.

Given that a finite number of projections must be performed, assessment of the results is often carried out in the form of ruin probability, that is, the outcomes are ranked in terms of a target measure (such as the shortfall of assets relative to liabilities at a specified future date).

Additional reserves are then set up at a level sufficient to cover all but a specified proportion of such shortfalls.

[5 Marks]

Solution 2:

- i) The globalization of capital markets has intensified the need for one set of financial reporting standards acceptable to all listing authorities in the major financial centres and which all investors use. IFRS Standards provides a high quality, internationally recognised set of accounting standards that bring transparency, accountability and efficiency to financial markets around the world.

This will make it easier for companies to obtain listings on different stock exchanges as they would not have to produce multiple sets of accounts.

It will also help investors making comparisons between companies listed on different exchanges.

Having consistent and reliable information helps investors to make informed choices, and hence reduces the risks they face. This reduction in risk should ultimately reduce the cost of capital to companies.

[4]

- ii) For institutions trading in investments, IFRS 9 requires:
- investments to be shown at fair value in the statement of financial position

- any resultant profit or loss on revaluation to be shown in the statement of comprehensive income

This has proved controversial, with institutions arguing that their investments are part of a longer-term portfolio and so their values should not be viewed on one particular day, with resultant fluctuations in the asset and income figures.

[3]

[7 Marks]

Solution 3:

i)

Term loans - fixed sum is lent to the borrower on a fixed date for a specified term.

Evergreen credit - permission to borrow up to a specified limit, with no fixed maturity.

So this is just like an overdraft facility, although it does involve the potential borrower paying a commitment fee to the bank for the option to borrow when it wants to.

Even with such an agreed line of evergreen credit, the bank may require the borrower to repay all outstanding amounts periodically to prevent what is intended to be a short-term form of borrowing becoming a long-term (or permanent) fixture.

Revolving credit (similar to evergreen credit, but with a fixed maturity of up to 3 years)

It is normal to have to pay a commitment fee on the full agreed amount of borrowing, whether or not it is all used. Interest is only payable on the amount actually borrowed.

Bridging loans (advances to be repaid from specified income) A bridging loan is used to pay for a specific item (eg investment in a new shop) in the interim period before long-term finance is obtained.

[4]

ii)

1. Commitment – whether there is prior commitment by the lender to advance funds when required (often requiring payment of a commitment fee to the lender).

Companies may just approach a bank for a loan “as and when” they need the cash. It is more usual, however, to pre-arrange a line of credit, ie to get a commitment from the bank on an amount up to which it will lend. This can

be on an evergreen or a revolving basis. A commitment fee is usually payable to the bank.

2. Maturity – the term for which the lending is made.
3. Rate of interest – this may be either fixed or floating. Very short-term loans may be at a fixed rate of interest. Most loans are at a floating rate, ie linked to the general level of interest rates, eg a fixed margin above LIBOR.
4. Security – whether the loan is to be secured against assets (either fixed or liquid assets).

[4]

iii)

$$L = 1,00,00,000$$

$$\delta_k = 0.50$$

$$F_k = 10\%$$

$$R_k = 12\%$$

$$r = 9\%$$

$$\sigma_k = 0.12$$

$$t_k = 1$$

$$P(0, \delta_{k+1}) = \text{Exp}(-r \times 1.5) = 0.8737$$

$$L \times \delta_k = 50,00,000$$

$$d_1 = -1.4594$$

$$d_2 = -1.5794$$

Therefore, value of option is

$$50,00,000 \times 0.8737 \times [0.12 N(-1.4594) - 0.10 N(-1.5794)]$$

$$= 1608.04$$

[6]

- iv) A swaption is an option to enter into an interest rate swap at a certain time in the future with a certain fixed rate being used. An interest rate swap can be regarded as the exchange of a fixed rate bond for a floating rate bond. A

swaption is therefore the option to exchange a fixed rate bond for a floating rate bond. The floating rate bond will be worth its face value at the beginning of the life of the swap. The swaption is therefore an option on a fixed rate bond with the strike price equal to the face value of the bond.

[4]

[18 Marks]**Solution 4:**

i)

a) The top-down approach to constructing and managing a portfolio involves a structured decision-making process which starts by considering the asset allocation at the highest level i.e. between different asset classes. Within each asset class an analysis is then made of how to distribute the available fund between different sectors (eg different industries for equities) and finally the selection of the individual assets to purchase is made.

A top-down investment strategy is therefore based upon deciding what the big picture is likely to be, before looking at the detail. The top-down approach would usually follow the steps set out below.

1. Decide upon the long-term benchmark or strategic allocation of assets between countries and between the main asset categories.
2. Decide on the short-term tactical split of investments, again between countries and between the main asset categories based on a shorter-term view of global economic and investment issues. The tactical decision may be based on fundamental, quantitative and technical analyses.
3. Given the chosen tactical asset allocation, decide upon the sector split within each asset category, eg whether banks will perform better than food retailers over the relevant time period within the domestic equity sector.
4. Finally, within each sector decide which particular stocks are “best value” (eg whether Bank X will outperform Bank Y within the banking sector). This will depend upon the results of an analysis of each individual company.

The actual buy and sell decisions with regard to individual investments are made on a day-to-day basis by the investment analysts working within each of the bond, equities and derivatives teams.

In contrast, a bottom-up approach seeks to identify the best value individual investments, irrespective of their geographic or sectoral spread.

The top-down approach lends itself better to controlling the risk of a portfolio by virtue of the fact that a balanced, diversified portfolio is held.

The main risk facing the investor is that of failing to meet its objectives, which in practice will often mean meeting its liabilities as they fall due. Continual reference to the investor's liabilities as part of a structured investment decision process ensures that the process focuses on the matching requirements of the fund, thereby reducing this risk.

Top-down adherents would also argue that the biggest differences in portfolio performance come from differences in asset allocation rather than in individual stock selection.

This may well depend on the markets considered. For example, currency movements will often be the biggest influence upon the domestic currency returns achieved from overseas investment, with stock selection less important. In contrast, stock selection within unquoted equities may be very important.

Concentration upon the big picture is a major advantage of the top-down approach, which means that it is most often used in practice. Note that although any specific institution will normally adopt one approach or the other, usually the top-down approach, there may still be scope for fund managers to achieve out-performance through "bottom-up", or individual stock selection.

It might be argued that starting with allocation between sectors ignores the fact that all investment performance starts with the performance of the individual assets held and that is where the analysis should be concentrated.

In other words, the whole is always simply the sum of the parts. However, the counter argument is that too much concentration on individual stocks will mean that less time is devoted to analysis of the bigger strategic issues, that will ultimately have a greater impact upon overall investment performance and the satisfactory achievement of investment objectives. For example, it may prove more difficult to attain the appropriate level of diversification via the bottom-up approach.

In practice it is often possible to develop a methodology which combines the two approaches.

For example, prior to each "top-down" meeting, each investment manager might be required to submit "buy" and "sell" recommendations within her own sector. A predominance of well-founded buy recommendations within a particular sector might then indicate that an increased allocation should be given to that sector. [10]

b) As a next project you have been asked to select between two investment trusts, X and Y, based on their risk adjusted performance. Trusts X and Y manage around 20% and 22%, respectively, of the total pension funds of ABC Company. You have been given the below information on the performance of these two trusts and the index over last 3 years.

	Trust X	Trust Y	Index
Mean return (%)	12	14	10
Beta	0.7	1.2	1
Standard deviation (%)	18	21	30

The mean risk-free return over the last 3 years is 8%.

[10]

- ii) As the investment trust(s) are held as a part of the pension fund's investments it is appropriate to focus on systematic risk only.

That is, we need to focus on portfolio beta measure – Treynor and Jensen.

Treynor measure

$$\text{Trust X} = (0.12 - 0.08) / 0.7 = 0.0571$$

$$\text{Trust Y} = (0.14 - 0.08) / 1.2 = 0.05$$

Jensen Measure

$$\text{Trust X} = 0.12 - [0.08 + 0.7 \times (0.10 - 0.08)] = 0.026$$

$$\text{Trust Y} = 0.14 - [0.08 + 1.2 \times (0.10 - 0.08)] = 0.036$$

On Teynor measure, Trust X outperforms Trust Y. However, on Jensen measure, Trust Y outperforms Trust X.

More information would be needed to reach a conclusion.

[6]

- iii)

The data is based on the last three years. There is no guarantee that the same relationships will hold in the future. For example, there may be changes in the way in which the investment trusts are run.

We are not told how the returns have been calculated (eg what allowance has been made for expenses, tax paid by the trusts, etc).

We have not considered the suitability of the trust for the fund's liabilities.

The Treynor and Jensen measures rest on the validity of the capital asset pricing Model.

[4]

[20 Marks]**Solution 5:**

- i) Exchange Traded Funds (ETFs) are the 'closed-ended' investment trust equivalents of (mutual) Index Funds. An ETF represents shares of ownership of a unit investment trust (UIT) which holds portfolios of stocks, bonds, currencies or commodities. The investor purchases the shares on a stock exchange in a process identical to the purchase or sale of any other listed stock.

[2]

- ii) ETFs are usually cheaper than index funds. But since they have to be purchased from the exchange the transaction costs have to be also considered when costs are compared

Index fund is available for purchase/sale at end of day net asset value. Index ETF can be purchased/sold during the market hours at prices close to real time index value.

When one purchases an index fund money is added to AUM of the fund. ETF is like a closed ended fund where you buy/sell when you have a counter party. To that extent availability of liquidity and low bid ask spread is critical for the success of an ETF.

Index funds usually maintain some cash to manage redemptions. This leads to tracking error. This is not the case with ETFs which doesn't have inflow/outflow of funds once it starts trading.

[5]

- iii) ETF is a passive asset management strategy. When fresh funds are diverted into ETFs it will lead to concentrated buying mainly in index stocks leading to index going up maybe much more than warranted relying only on flows. Conversely when flows reverse passive selling can lead to significant downfall in the market. ETF can accentuate volatility when stock market crisis happens.

Active funds usually evaluate the stocks that are bought and in a way ensure that price reflects the fundamentals. Passive funds don't bother to look into the fundamentals of the stock and price versus value distortion can be magnified as passive fund flows increase at the cost of active fund flows.

Active funds are usually significant stakeholders in companies and in a way play an important role in ensuring good corporate governance. If top stakeholders in a company

are ETFs then lack of good shareholders can lead to deterioration of corporate governance.

[6]

[13 Marks]

Solution 6:

i) Advantages

- An easy and elegant way to collect tax
- Increases transaction costs and reduces speculative activities like short term trading making markets more stable
- Unlike LTCG the tax collection depends on volume of transactions and hence tends to be stable irrespective of whether market goes up or down. Helps the government in fiscal planning in a better manner
- Attractiveness of Algaeland vis-à-vis other countries increases in the eyes of long term foreign investors if other countries levy LTCG

Disadvantages

- Short term traders provide liquidity. STT increases transaction costs which can lead to lower liquidity and increase frictional costs like higher bid ask spread
- STT has been introduced in lieu of capital gains – i.e. it is collected as an alternative to income tax. Market participants must pay STT irrespective of whether they make gains or losses which may not be considered fair
- Equity is just one form of asset class. Other classes like precious metals, real estate etc. will be taxed differently as LTCG maybe applicable here. This prevents creation of a level playing field between asset classes

[5]

ii) The following points maybe considered

- Investors were promised of STT replacing LTCG. The proposal implies both are to stay. This can lead to a market sell off
- Foreign investors may find other countries more attractive now. Capital may leave the country
- Markets have gone up sharply in the past few years. It maybe due for a correction. The previous month price as cost may lead to high cost levels which would imply LTCG will not be much after all in the event of a market correction. On the contrary it may lead to investors realizing capital loss which can be carried forward. Collections in the near term will not be much despite the disruption caused
- Lack of indexation benefit would mean that long term investors are severely disadvantaged especially considering the high level of inflation. If other asset classes enjoy indexation benefit then there will be criticism of different asset classes being taxed differently

- Market participants expect a consistency in taxation policy. Constant changes is not appreciated.

[5]

- iii) Suppose an investor has a 15-year time horizon. Before this proposal if he has bought the stock at 100 with a return expectation of 15%. If he is right the stock would appreciate to about 814 in the next 15 years. Under the new tax regime with 10% LTCG without indexation his net realization on this trade will be only 742 after paying the tax. So his annualized return will be only 14.3%. If he is unwilling to reduce his return expectation, then he will be willing to pay only about 91 for the same stock with similar expected appreciation instead of 100 earlier. {price to pay X will be obtained by solving the equation $[(814 - (814 - X) * 0.1) / X]^{(1/15)} = 15\%$ }. So for the same earnings the investor pays a lower price implying a lowering of P/E multiple.

[4]

- iv) The appreciation in stock price of a company over the long term is linked to the profits earned by the corporate and dividends paid by it. Companies pay tax on this profit before they are added to net worth. Since capital appreciation happens primarily out of post-tax earnings and dividends (which maybe attracting another tax) further taxation of the capital gains maybe deemed to be a case of multiple taxation.

[3]

- v) Any progressive tax regime requires that rich are taxed more than the poor. Having a same rate of 10% across the board goes against the principle of progressive taxation. Long term capital gains beyond a threshold only is taxed means that most of the investors of lower income group who get a small part of their annual income from long term capital gains remain unaffected. So number of tax payers affected by the change will be very less but the 10-20% of investors who account for say 80-90% of long term capital gains will only be taxed which is in line with principle of progressive taxation.

[2]

[19 Marks]**Solution 7:**

- i) The following needs to be considered and adjusted
- Ease of use
 - i. Readily available
 - ii. Objective
 - iii. Relevant for break up
 - iv. Auditable etc.
 - Book value is only one component of value. All else being equal lower P/BV can be considered more attractive. But everything else remaining the same is a very unlikely condition to be met
 - Book value is just an accounting number. It has to be adjusted for factors like investments/real estate held by company quoted in books at cost

- Qualitative parameters like relationships with customer, human skills (specialized areas like underwriting, actuarial etc.), geographic reach and distribution network and so on carry a lot of value which may not be fully reflected in book value
- Adjustments for companies that have expanded by acquisition – say for goodwill. Is it still relevant or should it be removed for making like to like comparison
- Value of a company is the present value of future free cash flows that the company can generate. Book value is a historic number
- Value depends not only book value but also how efficiently it is utilized – say what is the return on equity and how efficiently capital is utilized. So high ROE companies usually deserve higher P/BV while low ROE companies deserve lower P/BV

[6]

- ii) The required insurance profit margin will be about 6.2%
The working will be as follows

	Year 1	Year 2
Start Capital	36.07	40.00
Premium	100.00	120.00
Insurance Profit Margin	2%	6.2%
Profit	2.00	7.43
Investment Profit @ 10% of starting capital	3.61	4.00
PAT	5.61	11.43
RSM at 25% of premium	25.00	30.00
Dividend @ 30% of PAT	1.68	3.43
Surplus = PAT less dividend	3.93	8.00
Ending Capital	40.00	48.00
Solvency Ratio	1.60	1.60

Comments

- 20% is a sharp increase in turnover and getting it along with a higher insurance profit margin (nearly 3X the current margin) will be a tough ask
- With this kind of growth dividend policy maybe relooked into
- Scope to come down to 1.5x - some room for future growth

[6]

- iii) Important behavioral biases could include

- Over confidence – in his ability to predict when the market will change direction. Attributing his probable good performance in the rising market to his own skill versus luck

- Confirmation bias – fund manager will have an investment thesis and keeps looking for evidences that justify this rather than falsify this
- Hind sight bias – events that have happened are thought to be more predictable than they are prior to the actual event happening
- Anchoring – likelihood of an event happening are assessed based more on what has happened recently
- Dislike for negative event – over estimating the probability of events leading to positive outcomes for the fund
- Representative heuristics – assessing events easier to imagine as more probable
- Social proof – not to deviate too much from what fellow fund managers are doing

[6]

[18 Marks]
